CASO PRÁCTICO 3

Se pide

1 - Crear un archivo con los JSON siguientes:


```
{"Rk":1,"Name":"Michael
Jordan","Age":32,"G":82,"GS":82,"MP":37.7,"FG":11.2,"FGA":22.6,"FG%":0.
495,"3P":1.4,"3PA":3.2,"3P%":0.427,"2P":9.8,"2PA":19.4,"2P%":0.506,"eFG
%":0.525,"FT":6.7,"FTA":8,"FT%":0.834,"ORB":1.8,"DRB":4.8,"TRB":6.6,"AS
T":4.3,"STL":2.2,"BLK":0.5,"TOV":2.4,"PF":2.4,"PTS/G":30.4}

{"Rk":2,"Name":"Scottie
Pippen","Age":30,"G":77,"GS":77,"MP":36.7,"FG":7.3,"FGA":15.8,"FG%":0.4
63,"3P":1.9,"3PA":5.2,"3P%":0.374,"2P":5.4,"2PA":10.6,"2P%":0.507,"eFG%
":0.525,"FT":2.9,"FTA":4.2,"FT%":0.679,"ORB":2,"DRB":4.5,"TRB":6.4,"AST
":5.9,"STL":1.7,"BLK":0.7,"TOV":2.7,"PF":2.6,"PTS/G":19.4}

{"Rk":3,"Name":"Dennis
Rodman","Age":34,"G":64,"GS":57,"MP":32.6,"FG":2.3,"FGA":4.8,"FG%":0.48
,"3P":0,"3PA":0.4,"3P%":0.111,"2P":2.2,"2PA":4.3,"2P%":0.516,"eFG%":0.4
85,"FT":0.9,"FTA":1.7,"FT%":0.528,"ORB":5.6,"DRB":9.3,"TRB":14.9,"AST":
2.5,"STL":0.6,"BLK":0.4,"TOV":2.2,"PF":3.1,"PTS/G":5.5}

{"Rk":4,"Name":"Luc
Longley","Age":27,"G":62,"GS":62,"MP":26.5,"FG":3.9,"FGA":8.1,"FG%":0.4
82,"3P":0,"3PA":0,"3P%":0,"2P":3.9,"2PA":8.1,"2P%":0.482,"eFG%":0.482,"
FT":1.3,"FTA":1.7,"FT%":0.777,"ORB":1.7,"DRB":3.5,"TRB":5.1,"AST":1.9,"
STL":0.4,"BLK":1.4,"TOV":1.8,"PF":3.6,"PTS/G":9.1}

{"Rk":5,"Name":"Toni
Kukoc","Age":27,"G":81,"GS":20,"MP":26,"FG":4.8,"FGA":9.7,"FG%":0.49,"3
P":1.1,"3PA":2.7,"3P%":0.403,"2P":3.7,"2PA":7,"2P%":0.524,"eFG%":0.546,
"FT":2.5,"FTA":3.3,"FT%":0.772,"ORB":1.4,"DRB":2.6,"TRB":4,"AST":3.5,"S
TL":0.8,"BLK":0.3,"TOV":1.4,"PF":1.9,"PTS/G":13.1}

{"Rk":6,"Name":"Ron
Harper","Age":32,"G":80,"GS":80,"MP":23.6,"FG":2.9,"FGA":6.3,"FG%":0.46
7,"3P":0.4,"3PA":1.3,"3P%":0.269,"2P":2.6,"2PA":5,"2P%":0.519,"eFG%":0.
495,"FT":1.2,"FTA":1.7,"FT%":0.705,"ORB":0.9,"DRB":1.7,"TRB":2.7,"AST":
2.6,"STL":1.3,"BLK":0.4,"TOV":0.9,"PF":1.7,"PTS/G":7.4}

{"Rk":7,"Name":"Steve                                                Kerr
","Age":30,"G":82,"GS":0,"MP":23.4,"FG":3,"FGA":5.9,"FG%":0.506,"3P":1.
5,"3PA":2.9,"3P%":0.515,"2P":1.5,"2PA":3,"2P%":0.498,"eFG%":0.633,"FT":
1,"FTA":1,"FT%":0.929,"ORB":0.3,"DRB":1,"TRB":1.3,"AST":2.3,"STL":0.8,"
BLK":0,"TOV":0.5,"PF":1.3,"PTS/G":8.4}
```

{"Rk":8,"Name":"Bill Wennington","Age":32,"G":71,"GS":20,"MP":15,"FG":2.4,"FGA":4.8,"FG%":0.493,"3P":0,"3PA":0,"3P%":1,"2P":2.4,"2PA":4.8,"2P%":0.491,"eFG%":0.494,"FT":0.5,"FTA":0.6,"FT%":0.86,"ORB":0.8,"DRB":1.6,"TRB":2.5,"AST":0.6,"STL":0.3,"BLK":0.2,"TOV":0.5,"PF":2.4,"PTS/G":5.3}

{"Rk":9,"Name":"Dickey Simpkins","Age":23,"G":60,"GS":12,"MP":11.4,"FG":1.3,"FGA":2.7,"FG%":0.481,"3P":0,"3PA":0,"3P%":1,"2P":1.3,"2PA":2.7,"2P%":0.478,"eFG%":0.484,"FT":1,"FTA":1.6,"FT%":0.629,"ORB":1.1,"DRB":1.5,"TRB":2.6,"AST":0.6,"STL":0.2,"BLK":0.1,"TOV":0.9,"PF":1.3,"PTS/G":3.6}

{"Rk":10,"Name":"John Salley","Age":31,"G":17,"GS":0,"MP":11.2,"FG":0.7,"FGA":2.1,"FG%":0.343,"3P":0,"3PA":0,"3P%":0,"2P":0.7,"2PA":2.1,"2P%":0.343,"eFG%":0.343,"FT":0.7,"FTA":1.2,"FT%":0.6,"ORB":1.2,"DRB":1.4,"TRB":2.5,"AST":0.9,"STL":0.5,"BLK":0.9,"TOV":0.9,"PF":2.2,"PTS/G":2.1}

{"Rk":11,"Name":"Jud Buechler","Age":27,"G":74,"GS":0,"MP":10,"FG":1.5,"FGA":3.3,"FG%":0.463,"3P":0.5,"3PA":1.2,"3P%":0.444,"2P":1,"2PA":2.1,"2P%":0.474,"eFG%":0.545,"FT":0.2,"FTA":0.3,"FT%":0.636,"ORB":0.6,"DRB":0.9,"TRB":1.5,"AST":0.8,"STL":0.5,"BLK":0.1,"TOV":0.5,"PF":0.9,"PTS/G":3.8}

{"Rk":12,"Name":"Randy Brown","Age":27,"G":68,"GS":0,"MP":9.9,"FG":1.1,"FGA":2.8,"FG%":0.406,"3P":0,"3PA":0.2,"3P%":0.091,"2P":1.1,"2PA":2.7,"2P%":0.425,"eFG%":0.409,"FT":0.4,"FTA":0.7,"FT%":0.609,"ORB":0.3,"DRB":0.7,"TRB":1,"AST":1.1,"STL":0.8,"BLK":0.2,"TOV":0.5,"PF":1.3,"PTS/G":2.7}

{"Rk":13,"Name":"James Edwards","Age":40,"G":28,"GS":0,"MP":9.8,"FG":1.5,"FGA":3.9,"FG%":0.373,"3P":0,"3PA":0,"3P%":0,"2P":1.5,"2PA":3.9,"2P%":0.373,"eFG%":0.373,"FT":0.6,"FTA":0.9,"FT%":0.615,"ORB":0.5,"DRB":0.9,"TRB":1.4,"AST":0.4,"STL":0,"BLK":0.3,"TOV":0.8,"PF":2.2,"PTS/G":3.5}

{"Rk":14,"Name":"Jason Caffey","Age":22,"G":57,"GS":0,"MP":9.6,"FG":1.2,"FGA":2.8,"FG%":0.438,"3P":0,"3PA":0,"3P%":0,"2P":1.2,"2PA":2.8,"2P%":0.441,"eFG%":0.438,"FT":0.7,"FTA":1.2,"FT%":0.588,"ORB":0.9,"DRB":1.1,"TRB":1.9,"AST":0.4,"STL":0.2,"BLK":0.1,"TOV":0.8,"PF":1.6,"PTS/G":3.2}

{"Rk":15,"Name":"Jack Haley","Age":32,"G":1,"GS":0,"MP":7,"FG":2,"FGA":6,"FG%":0.333,"3P":0,"3PA":0,"3P%":0,"2P":2,"2PA":6,"2P%":0.333,"eFG%":0.333,"FT":1,"FTA":2,"FT%":0.5,"ORB":1,"DRB":1,"TRB":2,"AST":0,"STL":0,"BLK":0,"TOV":1,"PF":2,"PTS/G":5}

2 - Cargar este archivo en HDFS.

3 - Generar un *dataframe* desde el archivo que se acaba de subir a HDFS.

4 - Recuperar el esquema del *dataframe*.

5 - Recuperar los jugadores que promedian más de diez puntos por partido (PTS/G).

6 - Ordenar los jugadores por número de rebotes capturados (TRB).

7 - Crear una vista temporal con el comando *createOrReplaceTempView* y consultar las estadísticas de Michael Jordan.

8 - Obtener los jugadores que promediaron más de cinco rebotes por partido (TRB) y más de cinco asistencias por partido (AST).

9 - Ordenar la plantilla del equipo de mayor a menor edad (*age*).

10 - Calcular la media de edad del equipo.

Solución:

Antes de nada, lo primero que hacemos es comprobar las versiones de Java y Hadoop.

```
bigdata@bigdata-IMF:~$ java -version
openjdk version "1.8.0_312"
OpenJDK Runtime Environment (build 1.8.0_312-8u312-b07-0ubuntu1~20.04-b07)
OpenJDK 64-Bit Server VM (build 25.312-b07, mixed mode)
bigdata@bigdata-IMF:~$ hadoop version
Hadoop 3.3.1
Source code repository https://github.com/apache/hadoop.git -r a3b9c37a397ad4188041dd80621bdeefc46885f2
Compiled by ubuntu on 2021-06-15T05:13Z
Compiled with protoc 3.7.1
From source with checksum 88a4ddb2299aca054416d6b7f81ca55
This command was run using /home/bigdata/hadoop/share/hadoop/common/hadoop-common-3.3.1.jar
bigdata@bigdata-IMF:~$
```

Una vez configurado todo, iniciamos HDFS y YARN, y verificamos que los procesos necesarios se están ejecutando.

```
bigdata@bigdata-IMF:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bigdata-IMF]
bigdata@bigdata-IMF:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
bigdata@bigdata-IMF:~$ jps
47298 DataNode
49059 Jps
47525 SecondaryNameNode
48778 NodeManager
47146 NameNode
48299 ResourceManager
bigdata@bigdata-IMF:~$
```

1. Crear un archivo con los **JSON** siguientes:

    a. Creamos la carpeta ***spark_json*** y movemos el directorio a esta.

```
bigdata@bigdata-IMF:~$ mkdir -p ~/spark_json && cd ~/spark_json
bigdata@bigdata-IMF:~/spark_json$
```

    b. Creamos el archivo **JSON** y lo verificamos.

```
bigdata@bigdata-IMF:~/spark_json$ cat > bulls_96_97.jsonl <<'JSON'
> {"Rk":1,"Name":"Michael Jordan","Age":32,"G":82,"GS":82,"MP":37.7,"FG":11.2,"FGA":22.6,"FG%":0.49
34,"ORB":1.8,"DRB":4.8,"TRB":6.6,"AST":4.3,"STL":2.2,"BLK":0.5,"TOV":2.4,"PF":2.4,"PTS/G":30.4}
> {"Rk":2,"Name":"Scottie Pippen","Age":30,"G":77,"GS":77,"MP":36.7,"FG":7.3,"FGA":15.8,"FG%":0.463
679,"ORB":2,"DRB":4.5,"TRB":6.4,"AST":5.9,"STL":1.7,"BLK":0.7,"TOV":2.7,"PF":2.6,"PTS/G":19.4}
> {"Rk":3,"Name":"Dennis Rodman","Age":34,"G":64,"GS":57,"MP":32.6,"FG":2.3,"FGA":4.8,"FG%":0.48,"3
RB":5.6,"DRB":9.3,"TRB":14.9,"AST":2.5,"STL":0.6,"BLK":0.4,"TOV":2.2,"PF":3.1,"PTS/G":5.5}
> {"Rk":4,"Name":"Luc Longley","Age":27,"G":62,"GS":62,"MP":26.5,"FG":3.9,"FGA":8.1,"FG%":0.482,"3P
,"DRB":3.5,"TRB":5.1,"AST":1.9,"STL":0.4,"BLK":1.4,"TOV":1.8,"PF":3.6,"PTS/G":9.1}
> {"Rk":5,"Name":"Toni Kukoc","Age":27,"G":81,"GS":20,"MP":26,"FG":4.8,"FGA":9.7,"FG%":0.49,"3P":1.
.4,"DRB":2.6,"TRB":4,"AST":3.5,"STL":0.8,"BLK":0.3,"TOV":1.4,"PF":1.9,"PTS/G":13.1}
> {"Rk":6,"Name":"Ron  Harper","Age":32,"G":80,"GS":80,"MP":23.6,"FG":2.9,"FGA":6.3,"FG%":0.467,"3P
B":0.9,"DRB":1.7,"TRB":2.7,"AST":2.6,"STL":1.3,"BLK":0.4,"TOV":0.9,"PF":1.7,"PTS/G":7.4}
> {"Rk":7,"Name":"Steve Kerr ","Age":30,"G":82,"GS":0,"MP":23.4,"FG":3,"FGA":5.9,"FG%":0.506,"3P":1
"DRB":1,"TRB":1.3,"AST":2.3,"STL":0.8,"BLK":0,"TOV":0.5,"PF":1.3,"PTS/G":8.4}
> {"Rk":8,"Name":"Bill Wennington","Age":32,"G":71,"GS":20,"MP":15,"FG":2.4,"FGA":4.8,"FG%":0.493,"
8,"DRB":1.6,"TRB":2.5,"AST":0.6,"STL":0.3,"BLK":0.2,"TOV":0.5,"PF":2.4,"PTS/G":5.3}
> {"Rk":9,"Name":"Dickey Simpkins","Age":23,"G":60,"GS":12,"MP":11.4,"FG":1.3,"FGA":2.7,"FG%":0.481
.1,"DRB":1.5,"TRB":2.6,"AST":0.6,"STL":0.2,"BLK":0.1,"TOV":0.9,"PF":1.3,"PTS/G":3.6}
> {"Rk":10,"Name":"John Salley","Age":31,"G":17,"GS":0,"MP":11.2,"FG":0.7,"FGA":2.1,"FG%":0.343,"3P
DRB":1.4,"TRB":2.5,"AST":0.9,"STL":0.5,"BLK":0.9,"TOV":0.9,"PF":2.2,"PTS/G":2.1}
> {"Rk":11,"Name":"Jud Buechler","Age":27,"G":74,"GS":0,"MP":10,"FG":1.5,"FGA":3.3,"FG%":0.463,"3P"
":0.6,"DRB":0.9,"TRB":1.5,"AST":0.8,"STL":0.5,"BLK":0.1,"TOV":0.5,"PF":0.9,"PTS/G":3.8}
> {"Rk":12,"Name":"Randy Brown","Age":27,"G":68,"GS":0,"MP":9.9,"FG":1.1,"FGA":2.8,"FG%":0.406,"3P"
":0.3,"DRB":0.7,"TRB":1,"AST":1.1,"STL":0.8,"BLK":0.2,"TOV":0.5,"PF":1.3,"PTS/G":2.7}
> {"Rk":13,"Name":"James Edwards","Age":40,"G":28,"GS":0,"MP":9.8,"FG":1.5,"FGA":3.9,"FG%":0.373,"3
5,"DRB":0.9,"TRB":1.4,"AST":0.4,"STL":0,"BLK":0.3,"TOV":0.8,"PF":2.2,"PTS/G":3.5}
> {"Rk":14,"Name":"Jason Caffey","Age":22,"G":57,"GS":0,"MP":9.6,"FG":1.2,"FGA":2.8,"FG%":0.438,"3P
,"DRB":1.1,"TRB":1.9,"AST":0.4,"STL":0.2,"BLK":0.1,"TOV":0.8,"PF":1.6,"PTS/G":3.2}
> {"Rk":15,"Name":"Jack Haley","Age":32,"G":1,"GS":0,"MP":7,"FG":2,"FGA":6,"FG%":0.333,"3P":0,"3PA"
":0,"STL":0,"BLK":0,"TOV":1,"PF":2,"PTS/G":5}
> JSON
bigdata@bigdata-IMF:~/spark_json$
```

```
bigdata@bigdata-IMF:~/spark_json$ wc -l bulls_96_97.jsonl && head -n 2 bulls_96_97.jsonl
15 bulls_96_97.jsonl
{"Rk":1,"Name":"Michael Jordan","Age":32,"G":82,"GS":82,"MP":37.7,"FG":11.2,"FGA":22.6,"FG%":0.495,"3P
":1.4,"3PA":3.2,"3P%":0.427,"2P":9.8,"2PA":19.4,"2P%":0.506,"eFG%":0.525,"FT":6.7,"FTA":8,"FT%":0.834,
"ORB":1.8,"DRB":4.8,"TRB":6.6,"AST":4.3,"STL":2.2,"BLK":0.5,"TOV":2.4,"PF":2.4,"PTS/G":30.4}
{"Rk":2,"Name":"Scottie Pippen","Age":30,"G":77,"GS":77,"MP":36.7,"FG":7.3,"FGA":15.8,"FG%":0.463,"3P"
:1.9,"3PA":5.2,"3P%":0.374,"2P":5.4,"2PA":10.6,"2P%":0.507,"eFG%":0.525,"FT":2.9,"FTA":4.2,"FT%":0.679
,"ORB":2,"DRB":4.5,"TRB":6.4,"AST":5.9,"STL":1.7,"BLK":0.7,"TOV":2.7,"PF":2.6,"PTS/G":19.4}
bigdata@bigdata-IMF:~/spark_json$
```

2. Cargar este archivo en **HDFS**.

    a. Creamos el directorio en **HDFS**, subimos el archivo **JSON** y verificamos su existencia en dicha ruta.

```
bigdata@bigdata-IMF:~/spark_json$
bigdata@bigdata-IMF:~/spark_json$ hdfs dfs -mkdir -p /user/bigdata/nba
bigdata@bigdata-IMF:~/spark_json$ hdfs dfs -put -f bulls_96_97.jsonl /user/bigdata/nba/
bigdata@bigdata-IMF:~/spark_json$ hdfs dfs -ls /user/bigdata/nba/
Found 1 items
-rw-r--r--   1 bigdata supergroup       4266 2025-09-02 11:27 /user/bigdata/nba/bulls_96_97.jsonl
bigdata@bigdata-IMF:~/spark_json$
```

3. Generar un dataframe desde el archivo que se acaba de subir a **HDFS**.

    a. Abrimos **PySpark**, importamos todas sus funciones y leemos el **JSON** desde **HDFS**.

```
>>> from pyspark.sql import functions as F
>>> df = spark.read.json("hdfs:///user/bigdata/nba/bulls_96_97.jsonl")
>>>
```

4. Recuperar el esquema del dataframe.

```
>>> df.printSchema()
root
 |-- 2P: double (nullable = true)
 |-- 2P%: double (nullable = true)
 |-- 2PA: double (nullable = true)
 |-- 3P: double (nullable = true)
 |-- 3P%: double (nullable = true)
 |-- 3PA: double (nullable = true)
 |-- AST: double (nullable = true)
 |-- Age: long (nullable = true)
 |-- BLK: double (nullable = true)
 |-- DRB: double (nullable = true)
 |-- FG: double (nullable = true)
 |-- FG%: double (nullable = true)
 |-- FGA: double (nullable = true)
 |-- FT: double (nullable = true)
 |-- FT%: double (nullable = true)
 |-- FTA: double (nullable = true)
 |-- G: long (nullable = true)
 |-- GS: long (nullable = true)
 |-- MP: double (nullable = true)
 |-- Name: string (nullable = true)
 |-- ORB: double (nullable = true)
 |-- PF: double (nullable = true)
 |-- PTS/G: double (nullable = true)
 |-- Rk: long (nullable = true)
 |-- STL: double (nullable = true)
 |-- TOV: double (nullable = true)
 |-- TRB: double (nullable = true)
 |-- eFG%: double (nullable = true)

>>>
```

5. Recuperar los jugadores que promedian más de diez puntos por partido (PTS/G).

```
>>> spark.sql("""
...   SELECT Name, `PTS/G` AS PTS_per_G
...   FROM bulls
...   WHERE `PTS/G` > 10
...   ORDER BY `PTS/G` DESC
... """).show(truncate=False)
+--------------+---------+
|Name          |PTS_per_G|
+--------------+---------+
|Michael Jordan|30.4     |
|Scottie Pippen|19.4     |
|Toni Kukoc    |13.1     |
+--------------+---------+

>>>
```

6. Ordenar los jugadores por número de rebotes capturados (TRB).

```
>>> spark.sql("""
...   SELECT Name, TRB
...   FROM bulls
...   ORDER BY TRB DESC
... """).show(truncate=False)
+---------------+----+
|Name           |TRB |
+---------------+----+
|Dennis Rodman  |14.9|
|Michael Jordan |6.6 |
|Scottie Pippen |6.4 |
|Luc Longley    |5.1 |
|Toni Kukoc     |4.0 |
|Ron  Harper    |2.7 |
|Dickey Simpkins|2.6 |
|Bill Wennington|2.5 |
|John Salley    |2.5 |
|Jack Haley     |2.0 |
|Jason Caffey   |1.9 |
|Jud Buechler   |1.5 |
|James Edwards  |1.4 |
|Steve Kerr     |1.3 |
|Randy Brown    |1.0 |
+---------------+----+

>>>
```

7. Crear una vista temporal con el comando createOrReplaceTempView y consultar las estadísticas de Michael Jordan.

```
>>> df.createOrReplaceTempView("bulls")
>>>
>>> spark.sql("""
...   SELECT *
...   FROM bulls
...   WHERE Name = 'Michael Jordan'
... """).show(truncate=False)
```

```
>>> df.createOrReplaceTempView("bulls")
>>>
>>> spark.sql("""
...   SELECT *
...   FROM bulls
...   WHERE Name = 'Michael Jordan'
... """).show(truncate=False)
+---+-----+----+---+-----+---+---+---+---+---+----+-----+----+---+-----+---+---+---+----+--------------+---+---+-----+---+---+---+---+-----+
|2P |2P%  |2PA |3P |3P%  |3PA|AST|Age|BLK|DRB|FG  |FG%  |FGA |FT |FT%  |FTA|G  |GS |MP  |Name          |ORB|PF |PTS/G|Rk |STL|TOV|TRB|eFG% |
+---+-----+----+---+-----+---+---+---+---+---+----+-----+----+---+-----+---+---+---+----+--------------+---+---+-----+---+---+---+---+-----+
|9.8|0.506|19.4|1.4|0.427|3.2|4.3|32 |0.5|4.8|11.2|0.495|22.6|6.7|0.834|8.0|82 |82 |37.7|Michael Jordan|1.8|2.4|30.4 |1  |2.2|2.4|6.6|0.525|
+---+-----+----+---+-----+---+---+---+---+---+----+-----+----+---+-----+---+---+---+----+--------------+---+---+-----+---+---+---+---+-----+

>>>
```

8. Obtener los jugadores que promediaron más de cinco rebotes por partido (TRB) y más de cinco asistencias por partido (AST).

```
>>> spark.sql("""
...   SELECT Name, TRB, AST
...   FROM bulls
...   WHERE TRB > 5 AND AST > 5
...   ORDER BY TRB DESC, AST DESC
... """).show(truncate=False)
+--------------+---+---+
|Name          |TRB|AST|
+--------------+---+---+
|Scottie Pippen|6.4|5.9|
+--------------+---+---+

>>>
```

9. Ordenar la plantilla del equipo de mayor a menor edad (age).

```
>>> spark.sql("""
...   SELECT Name, Age
...   FROM bulls
...   ORDER BY Age DESC, Name
... """).show(truncate=False)
+---------------+---+
|Name           |Age|
+---------------+---+
|James Edwards  |40 |
|Dennis Rodman  |34 |
|Bill Wennington|32 |
|Jack Haley     |32 |
|Michael Jordan |32 |
|Ron  Harper    |32 |
|John Salley    |31 |
|Scottie Pippen |30 |
|Steve Kerr     |30 |
|Jud Buechler   |27 |
|Luc Longley    |27 |
|Randy Brown    |27 |
|Toni Kukoc     |27 |
|Dickey Simpkins|23 |
|Jason Caffey   |22 |
+---------------+---+

>>>
```

10. Calcular la media de edad del equipo.

```
>>> spark.sql("SELECT AVG(Age) AS avg_age FROM bulls").show()
+------------------+
|           avg_age|
+------------------+
|29.733333333333334|
+------------------+

>>>
```